

Aniruddhan Narasimhan

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Education

University of Maryland – College Park

M.S. in Applied Machine Learning

Aug 2025 – May 2027*

GPA: 3.7/4.0

Vellore Institute of Technology (VIT), Chennai

B.Tech. in Computer Science and Engineering (AI & Robotics)

2021 – May 2025

GPA: 3.6/4.0

Experience

Data Science Intern – KCF Technologies

Jun 2026 – Aug 2026

(Hybrid) State College, PA, USA

- Developed and evaluated **reinforcement learning-based fault classification models using PPO and DQN** across **30M+ industrial vibration sensor records**, multiple asset types, monitoring points, and X/Y vibration axes.
- Built scalable data preprocessing, experimentation, and inference workflows using **Python, PySpark, Databricks, and MLflow**, supporting missing-data analysis, feature preparation, asset-level validation, experiment tracking, and model inference.
- Identified and mitigated **temporal data leakage and class imbalance** across data splitting, imputation, and feature scaling, improving **unhealthy-class recall by 20–30%** while establishing a robust fault-classification pipeline evaluated using precision, recall, F1-score, and confusion matrices.

Trainee & Project Contributor – FincruX Technologies LLP

May 2024 – Aug 2024

Chennai, India

- Built predictive modeling pipelines for digital advertising data using **Python and statistical modeling**, improving predictive accuracy by **7%** through regularization, preprocessing, and data validation.
- Analyzed **time-series trends and advertising performance data** to identify revenue-impacting patterns supporting data-driven campaign and targeting decisions.

Research Experience

Research Author & Contributor – Vellore Institute of Technology, Chennai

Dec 2024 – Jun 2025

Dr. K. Valarmathi

- Co-authored **Enhanced Brain Tumour Prediction Using Quantum: A Hybrid Deep Learning Approach**, published in *Scientific Reports* (2026), and **Scalable Enhancement of Cloud-Based DDoS Detection with Adaptive ML Pipelines**, available on *TechRxiv*

Projects

Brain Tumor Classification | ResNet-50, TensorFlow, Keras, Scikit-learn, OpenCV



- Implemented and fine-tuned a **ResNet-50** image classification pipeline, improving accuracy by **12% over VGG-19** and achieving **99% AUC** for tumor versus non-tumor MRI classification.
- Improved model robustness through **data augmentation, class balancing, preprocessing, and threshold optimization**.

ALTA – Adaptive Logistics & Tracking Agent | LangGraph, LangChain, GPT-4o, FastAPI



- Built an **8-node LangGraph agentic AI pipeline** for pharmaceutical cold-chain monitoring with dynamic risk scoring and conditional routing. Developed a **3-stage autonomous rerouting agent** across **900+ cargo routes** with human-in-the-loop approval and compliance logging.

CompressIQ | Distributed Systems, Convex Optimization, PyTorch, CVXPY



- Designed **CompressIQ**, a bandwidth-aware gradient compression system for heterogeneous distributed training, achieving **1.85× faster simulated communication** than uniform compression across a **12-worker, 3-tier (10/5/1 Gbps) cluster** at matched test accuracy.
- Implemented a **CVXPY-based convex optimizer** with error-feedback-aware and per-layer compression for ring all-reduce, reducing communication time to **0.31× the uniform baseline** and achieving a cumulative **3.26× speedup** while maintaining **90.71% test accuracy**.

Geomagnetic Storm Early-Warning System | MLOps, AWS, FastAPI, MLflow, DVC



- Engineered an **end-to-end ML early-warning system** to forecast **Kp ≥ 5 geomagnetic storms 3 hours ahead**, training on **228K+ NASA OMNI2 hourly observations** with chronological validation, feature pipelines, schema checks, and MLflow-tracked classification experiments.
- Productionized **hourly inference from NOAA real-time solar-wind feeds** using DVC/S3 data versioning, FastAPI, Docker, and GitHub Actions CI/CD on AWS, with automated retraining and **Prometheus/Grafana/Evidently monitoring** for service health, data drift, and model performance.

Technical Skills

Programming: Production Python, SQL, R, C++.

Machine Learning: PySpark, Deep Learning, NLP, RAG, Time Series, Statistical Modeling, Model Optimization, MLOps.

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Seaborn, Streamlit, LangChain, LlamaIndex.

Tools & Platforms: Git, Docker, DVC, MLflow, GitHub Actions, FastAPI, Databricks, AWS, Azure, PostgreSQL, Microsoft SQL Server.